A PROJECT REPORT

on

"An Ensemble Hybrid Approach for Plant Disease Prediction Using Biosensor and Soil Data"

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN

INFORMATION TECHNOLOGY

BY

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March 2026

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CERTIFICATE

This is to certify that the project entitled

"An Ensemble Hybrid Approach for Plant Disease Prediction Using Biosensor and Soil Data"

submitted by

SREE HARSHITHA

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Information Technology) at KIIT Deemed to be University, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

Date: / /

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Project Guide

Acknowledgements

We are profoundly grateful to [GUIDE NAME] of School of Computer Engineering, KIIT University for expert guidance and continuous encouragement throughout to see that this project reaches its target since its commencement to its completion.

We also extend our gratitude to the faculty members of the School of Computer Engineering for their valuable support and resources provided during the course of this project.

Signature of Sree Harshitha

ABSTRACT

Plant diseases are a major threat to global agricultural productivity, causing an estimated 20-40% of annual crop losses worldwide. Traditional disease detection methods rely entirely on visual inspection, which identifies disease only after visible symptoms have already appeared on the plant — by which point significant damage has occurred and preventive action is no longer possible.

This project proposes an early warning system that predicts plant health status before symptoms appear, using continuous biosensor and soil sensor data. A hybrid ensemble model combining Random Forest and Support Vector Machine (SVM) via a stacking classifier is applied to a dataset of 1,200 readings recorded from 10 plants at 6-hour intervals over approximately one month. The dataset includes 11 environmental and soil features including Soil Moisture, Ambient Temperature, Humidity, Soil pH, Nitrogen, Phosphorus, Potassium, Chlorophyll Content, and Electrochemical Signal.

The proposed hybrid stacking model achieved ~98% prediction confidence, outperforming individual Random Forest and SVM classifiers. An additional Disease Risk Score (0-100) module was developed to quantify severity beyond class labels, and SHAP-based explainability was implemented to identify which sensor readings contributed most to each prediction. Results confirm that Chlorophyll Content and Electrochemical Signal are the strongest predictors of plant stress.

Keywords: Plant Disease Prediction, Random Forest, Support Vector Machine, Hybrid Ensemble, Biosensor Data, Disease Risk Score, Precision Agriculture, SHAP Explainability, Early Detection.

Chapter 1
Introduction

Plant diseases represent one of the most significant challenges to global food security and agricultural sustainability. According to the Food and Agriculture Organization (FAO), plant pathogens and pests are responsible for destroying 20-40% of global crop production annually, resulting in billions of dollars in economic losses and threatening the livelihoods of millions of farmers worldwide, particularly in developing agricultural economies such as India.

The conventional approach to plant disease detection relies heavily on visual inspection by trained agronomists or farmers. This method has a critical limitation: disease is only detectable after visible symptoms such as leaf discoloration, wilting, spots, or lesions have already manifested on the plant. By this stage, the pathogen has already established itself within the plant tissue, and in many cases, neighboring plants may already be exposed. This reactive approach makes large-scale prevention extremely difficult.

The emergence of Internet of Things (IoT) technology and biosensor systems has created an opportunity for a fundamentally different approach. Biosensors can continuously monitor environmental and soil conditions — temperature, humidity, soil moisture, pH, nutrient levels, chlorophyll concentration, and electrochemical stress signals — at high frequency and without human intervention. These parameters have well-established relationships with plant stress and disease susceptibility.

Machine learning algorithms, particularly ensemble methods such as Random Forest and kernel-based classifiers such as Support Vector Machine (SVM), have demonstrated strong performance on structured tabular data from sensor systems. By training these models on biosensor readings labeled with plant health status, it becomes possible to identify stress patterns before any visible symptoms appear.

This project proposes a hybrid ensemble model combining Random Forest and SVM via stacking, applied to biosensor and soil data for pre-symptomatic plant disease prediction. The main contributions are:

- A hybrid RF-SVM stacking ensemble model achieving ~98% prediction confidence

- A continuous Disease Risk Score (0-100) quantifying disease severity beyond class labels
- SHAP-based explainability identifying which sensor triggers each prediction
- Use of Electrochemical Signal and Chlorophyll Content as novel predictive features
- Three-way model comparison: Random Forest vs SVM vs Hybrid Stacking

The remainder of this report is structured as follows: Chapter 2 presents the literature review and basic concepts. Chapter 3 defines the problem statement and system design. Chapter 4 covers implementation, results, and analysis. Chapter 5 describes standards adopted. Chapter 6 concludes with future scope.

Chapter 2
Basic Concepts / Literature Review

2.1 Machine Learning for Plant Disease Detection

Machine learning has emerged as a powerful tool for agricultural applications. Supervised classification algorithms learn patterns from labeled training data and generalize to unseen samples. For plant disease prediction, the input features are sensor readings and the output labels are health status categories.

2.2 Random Forest

Random Forest, introduced by Breiman (2001), is an ensemble learning method that constructs multiple decision trees during training and outputs the class that is the mode of the classes (classification) of the individual trees. Each tree is built on a bootstrap sample of the training data, and at each split, a random subset of features is considered. This randomization reduces overfitting and improves generalization. Random Forest provides feature importance scores, making it highly interpretable for agricultural applications.

2.3 Support Vector Machine (SVM)

Support Vector Machine, proposed by Cortes and Vapnik (1995), is a supervised learning algorithm that finds the optimal hyperplane separating classes by maximizing the margin between the nearest data points (support vectors) of each class. The kernel trick — particularly the Radial Basis Function (RBF) kernel — enables SVM to handle non-linear decision boundaries by implicitly mapping data to higher-dimensional spaces. SVM is particularly effective on scaled, structured sensor data.

2.4 Hybrid Ensemble / Stacking

Stacking (stacked generalization) is an ensemble technique where predictions from multiple base models (Random Forest and SVM in this project) are used as input features to a meta-learner (Logistic Regression). This allows the meta-learner to learn how to best combine the base models' predictions, typically achieving higher accuracy than any individual model. The approach was formalized by Wolpert (1992).

2.5 Literature Review

Mohanty et al. (2016) applied deep convolutional neural networks (CNN) to the PlantVillage dataset of 54,306 leaf images and achieved 99.35% accuracy. However, this

approach requires high-quality cameras, significant compute resources, and can only detect disease after visible leaf symptoms appear — making it impractical for field-scale pre-symptomatic detection.

Ferentinos (2018) further explored CNN architectures for plant disease recognition via leaf images, confirming high accuracy but noting the same limitation: image-based models are inherently reactive rather than predictive.

Pantazi et al. (2019) proposed supervised machine learning using soil and crop sensor data for disease and stress detection, demonstrating that Random Forest achieves strong performance on structured tabular data from agricultural sensors.

Ahmed et al. (2021) compared SVM and Random Forest on datasets containing environmental parameters such as temperature, humidity, and soil nutrients. Both models achieved over 90% accuracy, with Random Forest outperforming SVM in multi-class scenarios.

Unlike all the above approaches, the proposed work uses a hybrid ensemble model on biosensor data including Electrochemical Signal and Chlorophyll Content — enabling automated, pre-symptomatic, real-time plant health prediction without requiring any visual inspection or imaging.

2.6 Key Tools and Technologies

Python 3.8+ was used as the primary programming language. Scikit-learn provided implementations of Random Forest, SVM, StackingClassifier, StandardScaler, and all evaluation metrics. Pandas and NumPy handled data manipulation. Matplotlib and Seaborn were used for visualization. Joblib enabled model persistence. Jupyter Notebook served as the development environment.

Chapter 3
Problem Statement / Requirement Specifications

Given continuous biosensor and soil sensor readings from plants, predict the plant health status (Healthy / Moderate Stress / High Stress) before visible disease symptoms appear, and quantify disease severity as a continuous risk score (0-100).

3.1 Project Planning

The project was executed in the following phases:

- Phase 1 — Data Collection and Exploration: Load and understand the biosensor dataset, check for missing values, analyze class distribution
- Phase 2 — Preprocessing: Drop non-predictive columns, encode labels, normalize features, split data
- Phase 3 — EDA: Generate visualizations including correlation heatmap, boxplots, and class distribution
- Phase 4 — Model Training: Train Random Forest, SVM, and Hybrid Stacking models
- Phase 5 — Evaluation: Compare models using accuracy, precision, recall, F1-score, and confusion matrix
- Phase 6 — Risk Score and Explainability: Generate Disease Risk Score and SHAP analysis
- Phase 7 — Prediction: Deploy saved model for real-time prediction on new sensor readings

3.2 Project Analysis

The dataset consists of 1,200 biosensor readings recorded from 10 plants at 6-hour intervals over approximately one month. After dropping Timestamp and Plant_ID, 11 numerical features remain. The target label Plant_Health_Status has three classes: Healthy (0), Moderate Stress (1), and High Stress (2). Initial analysis confirmed no missing values and sufficient class representation for supervised learning.

3.3 System Design

3.3.1 Design Constraints

Hardware: Standard laptop/desktop with Python 3.8+. No GPU required. Software: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib, Jupyter Notebook. Data: Biosensor-generated CSV dataset with 1,200 rows and 13 columns.

3.3.2 System Architecture / Block Diagram

The system follows a sequential ML pipeline:

- Step 1: Raw Sensor Data (CSV) → loaded via Pandas
- Step 2: Preprocessing → drop columns, encode labels, StandardScaler, 80/20 split
- Step 3: EDA → correlation heatmap, boxplots, class distribution
- Step 4: Base Models → Random Forest (n=100) + SVM (RBF kernel) trained independently
- Step 5: Hybrid Stacking → RF + SVM predictions fed into Logistic Regression meta-learner
- Step 6: Evaluation → accuracy, confusion matrix, classification report, cross-validation
- Step 7: Risk Score → continuous 0-100 severity score from model probabilities
- Step 8: SHAP Explainability → feature contribution analysis
- Step 9: Prediction → load .pkl model, scale input, output health status + confidence

Chapter 4
Implementation

4.1 Methodology / Proposal

The project implements a complete machine learning pipeline across 9 Jupyter notebooks. Each notebook handles a distinct phase of the pipeline, making the code modular and reproducible.

4.1.1 Data Preprocessing

Timestamp and Plant_ID columns were dropped as they carry no predictive value. Labels were encoded using a manual mapping: Healthy=0, Moderate Stress=1, High Stress=2. Feature scaling was applied using StandardScaler (mean=0, std=1) to normalize the vastly different ranges of sensor readings. The dataset was split 80% training (960 samples) and 20% testing (240 samples) using stratified sampling.

4.1.2 Random Forest Model

Random Forest was configured with n_estimators=100, random_state=42, and n_jobs=-1 for parallel processing. The model was trained on the scaled training set and evaluated on the test set using accuracy score, classification report, and confusion matrix. Feature importance scores were extracted and visualized. Cross-validation (5-fold) confirmed generalization.

4.1.3 SVM Model

SVM was configured with kernel='rbf', C=1.0, gamma='scale', and probability=True (required for soft voting and stacking). The model was trained on the same scaled data and evaluated with identical metrics.

4.1.4 Hybrid Stacking Model (Novelty)

The hybrid model uses Scikit-learn's StackingClassifier with Random Forest and SVM as base estimators and Logistic Regression as the meta-learner. The meta-learner is trained on out-of-fold predictions from the base models using 5-fold cross-validation, preventing data leakage. This approach allows the meta-learner to learn the optimal weighting of RF and SVM predictions, achieving superior accuracy over either individual model.

4.1.5 Disease Risk Score

A continuous risk score (0-100) was derived from the model's probability outputs: Risk Score = (P(Moderate Stress) x 50) + (P(High Stress) x 100). This converts the probabilistic output into an actionable severity index. Score ranges: 0-30 (Safe), 31-60 (Monitor), 61-80 (Warning), 81-100 (Critical).

4.2 Testing / Verification Plan

Test ID	Test Case	Condition	System Behavior	Expected Result
T01	Healthy Prediction	Soil 60%, Humidity 45%, Chlorophyll 38	Model predicts Healthy	Healthy, >95% conf
T02	High Stress Prediction	Soil 15%, Humidity 88%, Chlorophyll 12	Model predicts High Stress	High Stress, >95% conf
T03	Risk Score	High Stress sample	Risk score computed	Score > 75
T04	Batch Prediction	2 samples input	Both predicted correctly	Healthy + High Stress
T05	Model Saving	After training	PKL files saved	Files exist in /models

4.3 Result Analysis

The following table summarizes the performance of all three models on the test set:

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	~96%	High	High	High
SVM (RBF)	~93%	High	Moderate	Moderate
Hybrid Stacking	~98%	Highest	Highest	Highest

Key findings from result analysis: The Hybrid Stacking model outperformed both individual models. Prediction confidence on new sensor readings reached 98.51% (Healthy) and 98.85% (High Stress) as verified in notebook 08_prediction.ipynb. Chlorophyll Content and Electrochemical Signal were identified as the strongest predictors of plant stress via feature importance analysis.

4.4 Quality Assurance

Quality assurance was maintained through: (1) 5-fold cross-validation to prevent overfitting, (2) stratified train-test split ensuring balanced class representation, (3) notebook-based reproducible code with fixed random seeds (random_state=42), (4) version-controlled codebase on GitHub, and (5) modular pipeline with independent notebooks enabling individual step verification.

Chapter 5
Standards Adopted

5.1 Design Standards

The project follows IEEE standards for machine learning system design. The data pipeline adheres to the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. UML-style flowcharts were used to document the system architecture.

5.2 Coding Standards

The following coding standards were adhered to throughout the project:

- PEP 8 Python style guide for code formatting and naming conventions
- Descriptive variable names (e.g., X_train, y_test, hybrid_model) for readability
- Functions are modular — each notebook handles a single pipeline stage
- All code is commented with clear section headers
- Fixed random seeds (random_state=42) for reproducibility
- Warning suppression (warnings.filterwarnings) for clean output
- File paths handled with pathlib.Path for cross-platform compatibility

5.3 Testing Standards

Testing followed IEEE 829 standard for software test documentation. Model validation used k-fold cross-validation (k=5) as recommended by best practices for machine learning evaluation. All evaluation metrics (accuracy, precision, recall, F1-score) follow standard definitions from the machine learning community. Test cases were defined prior to implementation and verified against expected results.

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

This project successfully developed an end-to-end machine learning pipeline for pre-symptomatic plant disease prediction using biosensor and soil data. The proposed hybrid ensemble model combining Random Forest and SVM via stacking achieved approximately 98% prediction confidence, outperforming both individual models.

The system introduces four novel contributions to the field: (1) pre-symptomatic detection using sensor data before visible symptoms appear, (2) a hybrid RF-SVM stacking ensemble as the core classifier, (3) a continuous Disease Risk Score (0-100) for severity quantification, and (4) SHAP-based explainability identifying which sensor readings trigger each prediction. Chlorophyll Content and Electrochemical Signal were confirmed as the strongest predictors of plant stress.

The approach is practically superior to image-based methods as it requires no cameras, operates continuously via IoT sensors, and provides actionable early warnings before irreversible crop damage occurs. The saved model files (.pkl) are deployment-ready for integration into web services or IoT edge devices.

6.2 Future Scope

Several directions exist for extending this work:

- Expand the dataset to include more plant species, crop types, and geographic regions for better generalization
- Apply LSTM or Transformer models on time-series sensor sequences to capture temporal stress patterns
- Develop a mobile application for farmers with real-time alerts integrated with IoT biosensors
- Deploy on Raspberry Pi or Arduino hardware with physical soil and humidity sensors for live field monitoring
- Extend from stress level classification to specific disease identification (fungal, bacterial, viral)
- Investigate federated learning approaches for privacy-preserving model training across multiple farms

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